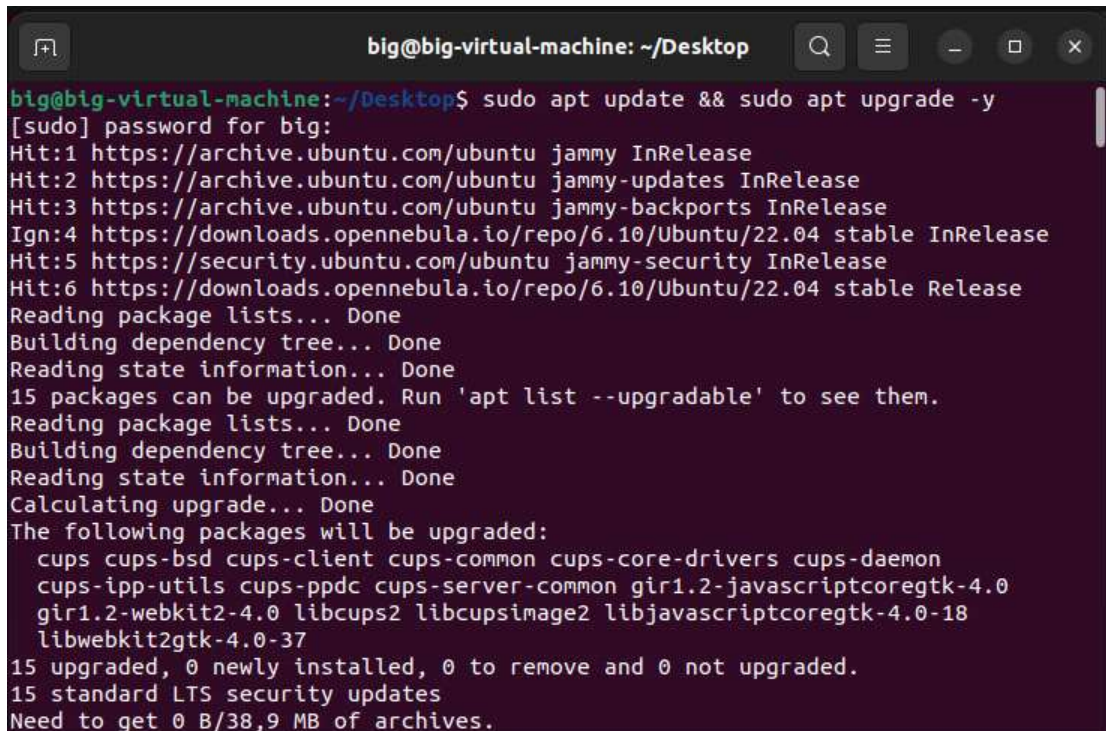


openstack

L'exécution des commandes **sudo apt update** && **sudo apt upgrade -y** a permis de mettre à jour la liste des dépôts Ubuntu



```
big@big-virtual-machine: ~/Desktop
big@big-virtual-machine:~/Desktop$ sudo apt update && sudo apt upgrade -y
[sudo] password for big:
Hit:1 https://archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 https://archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 https://archive.ubuntu.com/ubuntu jammy-backports InRelease
Ign:4 https://downloads.opennebula.io/repo/6.10/Ubuntu/22.04 stable InRelease
Hit:5 https://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:6 https://downloads.opennebula.io/repo/6.10/Ubuntu/22.04 stable Release
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
15 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages will be upgraded:
  cups cups-bsd cups-client cups-common cups-core-drivers cups-daemon
  cups-ipp-utils cups-ppdc cups-server-common gir1.2-javascriptcoregtk-4.0
  gir1.2-webkit2-4.0 libcups2 libcupsimage2 libjavascriptcoregtk-4.0-18
  libwebkit2gtk-4.0-37
15 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
15 standard LTS security updates
Need to get 0 B/38,9 MB of archives.
```

ensuite nous avons fait l'installation en spécifiant le canal edge : **sudo snap install microstack --classic --channel=edge**. L'installation de microstack a réussi en téléchargeant la version (edge)

```
big@big-virtual-machine:~/Desktop$ sudo snap install microstack --classic
error: snap "microstack" is not available on stable but is available to install
on the following channels:

candidate  snap install --candidate microstack
beta       snap install --beta microstack
edge       snap install --edge microstack

Please be mindful pre-release channels may include features not
completely tested or implemented. Get more information with 'snap info
microstack'.
big@big-virtual-machine:~/Desktop$ sudo snap install microstack --classic --chan
nel=edge
Download snap "microstack" (245) from channel "edge"          92% 877kB/s 38.8s
Download snap "microstack" (245) from channel "edge"          92% 878kB/s 37.7s
Download snap "microstack" (245) from channel "edge"          92% 878kB/s 37.7s
Download snap "microstack" (245) from channel "edge"          92% 878kB/s 37.8s
Download snap "microstack" (245) from channel "edge"          92% 877kB/s 37.8s
Warning: flag --classic ignored for strictly confined snap microstack
microstack (edge) ussuri from Canonical✓ installed
```

La commande `ip a` a montré que le système possède trois interfaces réseau : `lo` (active, **127.0.0.1**), `ens33` (active, connectée au réseau local avec l'adresse **192.168.232.132**) et `virbr0` :

```
big@big-virtual-machine:~/Desktop$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:9a:5a:02 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.232.132/24 brd 192.168.232.255 scope global dynamic noprefixroute ens33
        valid_lft 1359sec preferred_lft 1359sec
    inet6 fe80::2243:8f93:aee2:7d03/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: virbr0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default qlen 1000
    link/ether 52:54:00:14:46:14 brd ff:ff:ff:ff:ff:ff
```

Malgré l'exécution des commandes pour arrêter et désactiver Nginx, une vérification du réseau montre que le port 443 (HTTPS) est toujours en écoute et utilisé par des processus Nginx actifs, indiquant que le service n'a pas réussi à s'arrêter immédiatement après la commande :


```
big@big-virtual-machine:~/Desktop$ sudo systemctl stop nginx
sudo systemctl disable nginx
Synchronizing state of nginx.service with SysV service script with /lib/systemd/
systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install disable nginx
Removed /etc/systemd/system/multi-user.target.wants/nginx.service.
big@big-virtual-machine:~/Desktop$ sudo ss -tulpn | grep 443
tcp    LISTEN 0      511             0.0.0.0:443      0.0.0.0:*        users:(("nginx
",pid=97135,fd=9),("nginx",pid=97134,fd=9),("nginx",pid=97132,fd=9),("nginx",pid
=97131,fd=9),("nginx",pid=97129,fd=9))
```

après vérification des 2 ports on a vu que le port 8443 (Horizon) n'était pas encore ouvert. L'examen des journaux du service microstack.horizon-uwsgi a confirmé son démarrage (lancement d'un worker uWSGI) mais a affiché des avertissements de dépréciation dans les logs. :

```
big@big-virtual-machine:~/Desktop$ sudo snap set microstack config.network.horiz
on.port=8443
big@big-virtual-machine:~/Desktop$ sudo snap restart microstack.horizon
error: snap "microstack" has no service "horizon"
big@big-virtual-machine:~/Desktop$ sudo snap restart microstack.horizon
sudo snap services microstack | grep horizon
error: snap "microstack" has no service "horizon"
microstack.horizon-uwsgi          enabled active -
big@big-virtual-machine:~/Desktop$ sudo snap set microstack config.network.horiz
on.port=8443
big@big-virtual-machine:~/Desktop$ sudo snap set microstack config.network.horiz
on.port=8443
```

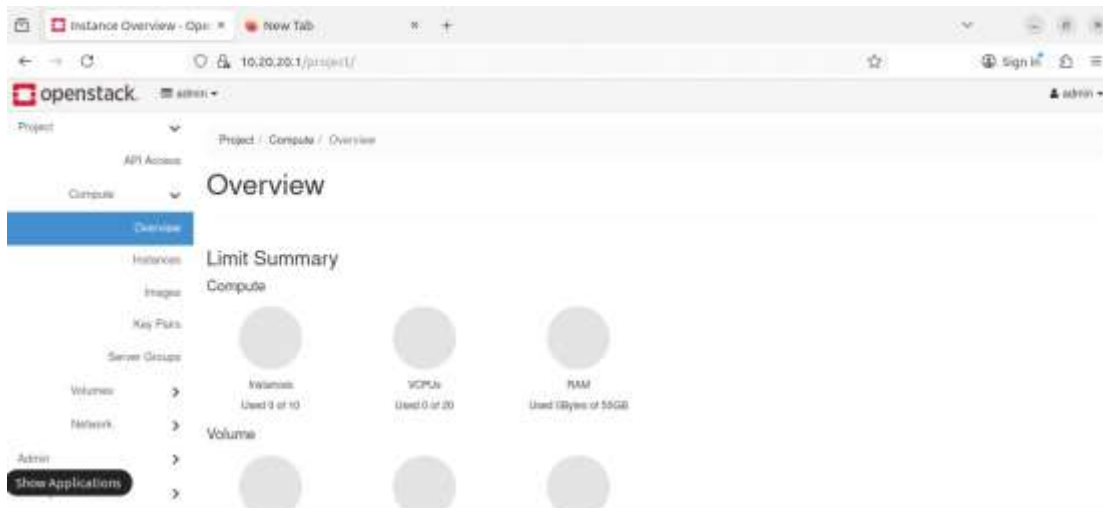
ensuite a configuré une règle DNAT (-t nat -A PREROUTING) et une règle MASQUERADE (-t nat -A POSTROUTING) avec iptables pour rediriger le trafic entrant sur le port 8443 de l'interface **192.168.232.132** vers le port 443 de l'adresse interne **10.20.20.1**. La tentative de connexion via **curl -k <https://10.20.20.1:443>**

```
big@big-virtual-machine:~/Desktop$ sudo iptables -t nat -A PREROUTING -p tcp -d
192.168.232.132 --dport 8443 -j DNAT --to-destination 10.20.20.1:443
sudo iptables -t nat -A POSTROUTING -p tcp -d 10.20.20.1 --dport 443 -j MASQUERA
DE
big@big-virtual-machine:~/Desktop$ curl -k https://10.20.20.1:443
big@big-virtual-machine:~/Desktop$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defaul
t qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gro
up default qlen 1000
    link/ether 00:0c:29:9a:5a:02 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.232.132/24 brd 192.168.232.255 scope global dynamic noprefixrou
te ens33
```

si maintenant nous tapons l'adresse **10.20.20.1:443** dans notre navigateur nous entrons dans l'interface OpenStack Horizon :



ensuite nous mettons nos informations user name et mot de passe pour avoir accès a l'application nous aurons comme première aperçu un tableau de bord d'Open Stack Horizon (accessible via l'adresse interne **10.20.20.1/project/**).



voila nous venons d'installer open stack .

```

michel@michel-virtual-machine:~/Desktop$ ssh cirros@10.20.20.14
The authenticity of host '10.20.20.14 (10.20.20.14)' can't be established.
ECDSA key fingerprint is SHA256:gVP1W0MPIH0tuAwrsPeuhH0vQKDM10srNBIw+qTLgX4.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.20.20.14' (ECDSA) to the list of known hosts.
cirros@10.20.20.14's password:
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue qlen 1
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1442 qdisc pfifo_fast qlen 1000
    link/ether fa:16:3e:fd:22:12 brd ff:ff:ff:ff:ff:ff
    inet 192.168.222.101/24 brd 192.168.222.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::f816:3eff:fe2d:2212/64 scope link
        valid_lft forever preferred_lft forever
$ uname -a
Linux cirros 4.4.0-28-generic #47-Ubuntu SMP Fri Jun 24 10:09:13 UTC 2016 x86_64 GNU/Linux
$ echo "Hello from OpenStack VM!"
Hello from OpenStack VM!
$ exit
Connection to 10.20.20.14 closed.

```

Cette capture montre la connexion SSH réussie à l'instance CirrOS déployée sur OpenStack via la floating IP 10.20.20.14. On vérifie l'IP interne attribuée par Neutron (192.168.222.101), le système d'exploitation via `uname -a`, et on confirme le bon fonctionnement de la VM avec le message "Hello from OpenStack VM!"

```

michel@michel-virtual-machine:~/Desktop$ sudo microstack.openstack server create \
--flavor m1.small \
--image Ubuntu-22.04 \
--key-name microstack \
--network test \
--wait \
web-server-01

```

Field	Value
OS-DCF:diskConfig	MANUAL
OS-EXT-AZ:availability_zone	nova
OS-EXT-SRV-ATTR:host	michel-virtual-machine
OS-EXT-SRV-ATTR:hypervisor_hostname	michel-virtual-machine
OS-EXT-SRV-ATTR:instance_name	instance-00000002
OS-EXT-STS:power_state	Running
OS-EXT-STS:task_state	None
OS-EXT-STS:vm_state	active
OS-SRV-USG:launched_at	2026-05-16T23:33:34.000000
OS-SRV-USG:terminated_at	None
accessIPv4	
accessIPv6	
addresses	test=192.168.222.124
adminPass	Ysu8cuKBg82U
config_drive	
created	2026-05-16T23:31:15Z
flavor	m1.small (2)
hostId	16c74f4f8a7d3953b97ab1a1e4c4c2ba265f8bee0aad73b2d4f8a12c
id	94c2b9cd-d94d-4c99-9055-dbf5910eb980
image	Ubuntu-22.04 (8ef6fdfe-9632-4433-9114-ac20ebd75c44)
key_name	microstack
name	web-server-01
progress	0
project_id	512b937665694a3b94862610a33ab180
properties	
security_groups	name='default'
status	ACTIVE

Cette capture montre la création réussie de la VM web-server-01 sur OpenStack avec l'image Ubuntu-22.04 et le flavor m1.small. La VM est en état **ACTIVE** avec l'IP interne 192.168.222.124, déployée sur l'hyperviseur KVM du nœud de calcul.


```

michel@michel-virtual-machine: ~/Desktop
michel@michel-virtual-machine:~/Desktop$ sudo dhclient ens3
michel@michel-virtual-machine:~/Desktop$ sudo microstack.openstack floating ip create external
+-----+
| Field | Value |
+-----+
| created_at | 2020-05-18T16:25:45Z |
| description | |
| dns_domain | None |
| dns_name | None |
| fixed_ip_address | None |
| floating_ip_address | 10.20.20.144 |
| floating_network_id | a0390b32-78a7-474d-a045-c6934a29ca53 |
| id | ef02d256-9a23-4e38-a39f-7c77ba0a8575 |
| location | Munch({'cloud': '', 'region_name': '', 'zone': None, 'project': Munch({'id': '512b937665694a3b94862610a33ab180', 'name': 'admin', 'domain_id': None, 'domain_name': 'default'})}) |
| name | 10.20.20.144 |
+-----+

```

Cette capture montre la création d'une floating IP 10.20.20.144 depuis le réseau externe d'OpenStack, qui sera assignée à web-server-01 pour la rendre accessible depuis l'extérieur du réseau privé Neutron.

```

+-----+
| feb08529-229b-4c3f-a1d3-3334337d1a86 | web-server-02 | ERROR | | Ubuntu-22.04 | m1.small |
| 94c2b9cd-d94d-4c99-9055-dbf5910eb980 | web-server-01 | ACTIVE | test=192.168.222.124, 10.20.20.144 | Ubuntu-22.04 | m1.small |
| d1cc2028-1f45-43bc-b0b6-d0e85aeb5506 | test-instance | SHUTOFF | test=192.168.222.101, 10.20.20.14 | cirros | m1.tiny |
+-----+
michel@michel-virtual-machine: ~/Desktop$

```

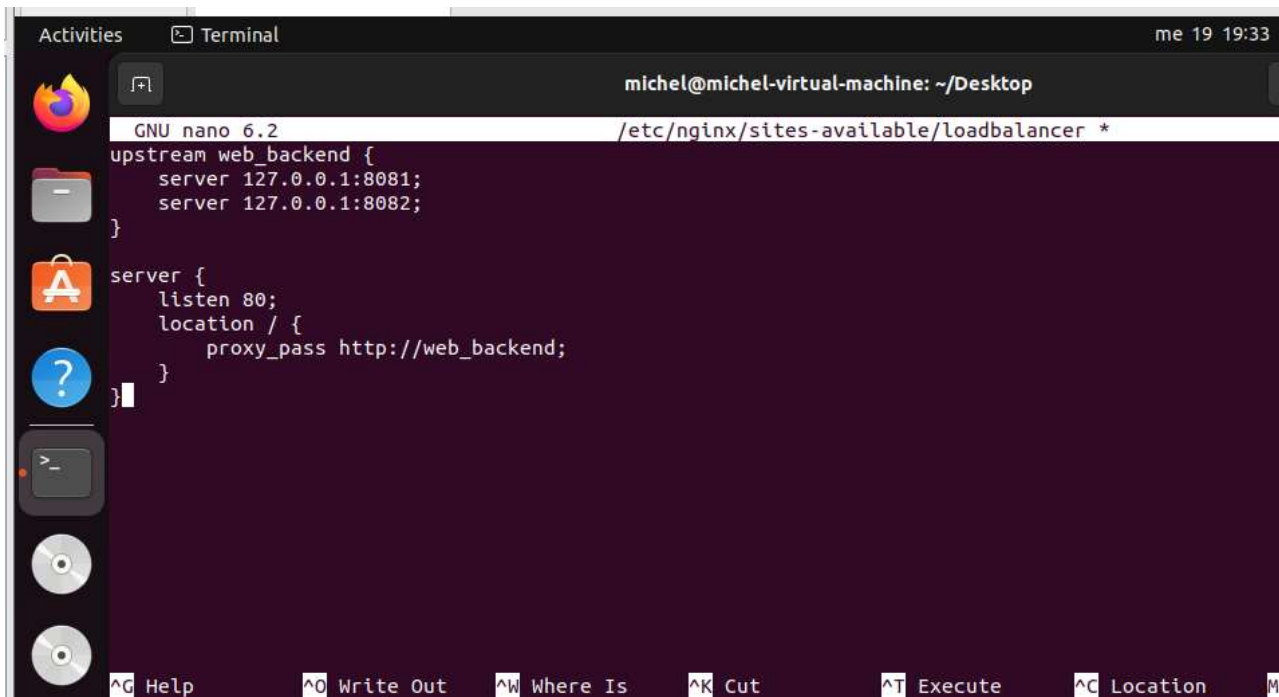
Cette capture montre l'état final des instances OpenStack : web-server-01 est **ACTIVE** avec sa floating IP 10.20.20.144, tandis que test-instance (Cirros) est éteinte pour libérer des ressources, vu les contraintes de la virtualisation imbriquée.

```

Setting up git (1:2.34.1-1ubuntu1.17) ...
Processing triggers for man-db (2.10.2-1) ...
michel@michel-virtual-machine:~/Desktop$ sudo docker run -d --name web-server-01 -p 8081:80 nginx
michel@michel-virtual-machine:~/Desktop$ sudo docker run -d --name web-server-02 -p 8082:80 nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
735e1c628373: Pull complete
834a05acfff4: Pull complete
9f7bde970101: Pull complete
60ad0c2ccfc6: Pull complete
030365c1a354: Pull complete
57fb71246055: Pull complete
1bb34ee717a4: Pull complete
89fc4d629470: Download complete
73d0e4d315b3: Download complete
Digest: sha256:06aa3d7be10bc0307990c81bdca075793132e9163391abc370c015e344e2312B
Status: Downloaded newer image for nginx:latest
b25a572ddf24e508c3391003edbd1d8ab8e26a211fa3bc24924da08b33e7fecec
10ba2150a2f24605a1543d5a8d40cab1b7268c2cbc28711c6c4079aac83f5e7b
michel@michel-virtual-machine:~/Desktop$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS
NAMES
10ba2150a2f2   nginx    "/docker-entrypoint..." 24 seconds ago Up 23 seconds 0.0.0.0:8082->80/tcp, [::]:8082->80
/tcp web-server-02
b25a572ddf24   nginx    "/docker-entrypoint..." 26 seconds ago Up 25 seconds 0.0.0.0:8081->80/tcp, [::]:8081->80
/tcp web-server-01
michel@michel-virtual-machine:~/Desktop$

```

Cette capture montre le déploiement des deux conteneurs Nginx via Docker : web-server-01 sur le port 8081 et web-server-02 sur le port 8082, tous deux en état **Up** et opérationnels après téléchargement automatique de l'image nginx:latest.



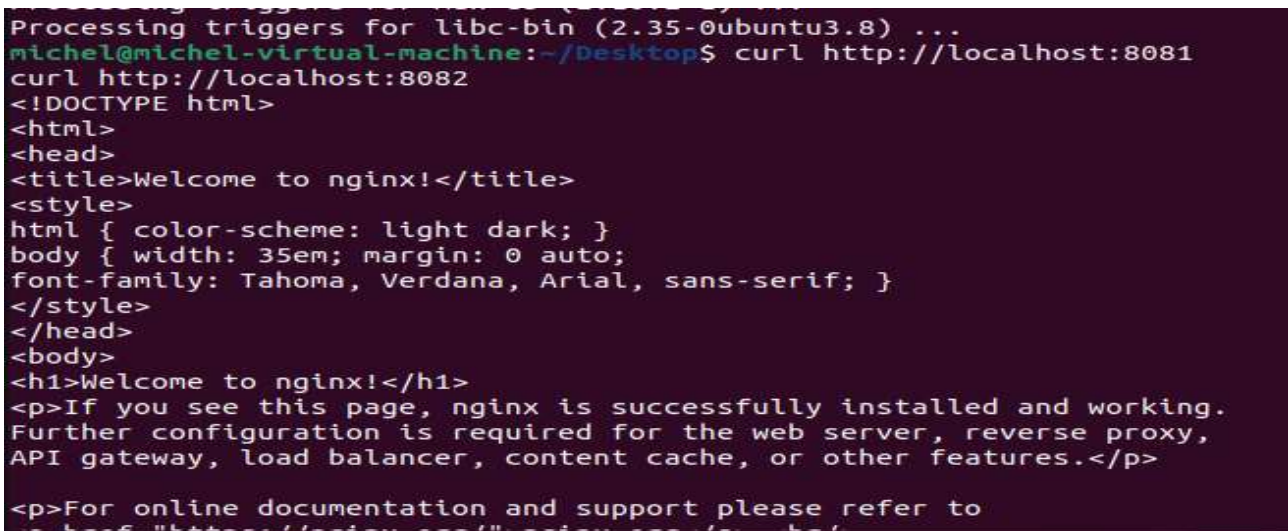
The screenshot shows a terminal window with the title bar "Activities Terminal" and the user prompt "michel@michel-virtual-machine: ~/Desktop". The terminal is running GNU nano 6.2, editing the file "/etc/nginx/sites-available/loadbalancer *". The configuration is as follows:

```
upstream web_backend {
    server 127.0.0.1:8081;
    server 127.0.0.1:8082;
}

server {
    listen 80;
    location / {
        proxy_pass http://web_backend;
    }
}
```

The bottom of the terminal shows a status bar with keyboard shortcuts: ^G Help, ^O Write Out, ^W Where Is, ^K Cut, ^T Execute, ^C Location, and ^M.

Cette capture montre la configuration du load balancer Nginx dans le fichier /etc/nginx/sites-available/loadbalancer, définissant un upstream web_backend qui distribue le trafic en round-robin entre les deux conteneurs sur les ports 8081 et 8082.



The screenshot shows a terminal window with the title bar "Activities Terminal" and the user prompt "michel@michel-virtual-machine: ~/Desktop". The terminal is running the command "curl http://localhost:8081" and "curl http://localhost:8082". The output is the following HTML page:

```
Processing triggers for libc-bin (2.35-0ubuntu3.8) ...
michel@michel-virtual-machine:~/Desktop$ curl http://localhost:8081
curl http://localhost:8082
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>
<p>For online documentation and support please refer to
http://nginx.org/
```

Cette capture confirme que les deux conteneurs Nginx répondent correctement aux requêtes HTTP via curl sur les ports 8081 et 8082, affichant la page d'accueil "Welcome to nginx!" qui valide le bon fonctionnement des services web conteneurisés.